

Weekly Self-Study Tasks to Complete

React Composition and Hooks for Leaders

This document covers all of Week 4 of Phase 2 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen Insurance Quote Page lab, where you will refactor the React app with a custom hook and the Context API. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~60 min)

Watch the specific resources called out below — not an entire course. Keep the running app open so you can connect custom hooks, context, and reducers to the Evergreen refactor.

Topic	Material Description	Est. Time	Link / Reference
Hooks in Practice	Using React 17 Hooks (Peter Kellner, Pluralsight, updated Jun 26, 2024; 3h 21m course). Watch “Start Using React Hooks with useState, useEffect, and useRef” (~28 min of the 44-min module). Shows the core hooks in real functional components. Reinforces the instructor demo.	~28 min	pluralsight.com — Using React 17 Hooks
Custom Hooks	React.dev “Reusing Logic with Custom Hooks” (official docs) — the guide to extracting stateful logic into a use-prefixed function, with runnable examples.	~17 min	react.dev/learn — Custom Hooks
Context API	React.dev “Passing Data Deeply with Context” (official docs) — read and run the examples on the	~15 min	react.dev/learn — Context

Topic	Material Description	Est. Time	Link / Reference
	provider/consumer and the reducer + context pattern.		

Practice Activities (1 hour)

Complete this week's hands-on lab: "Evergreen Insurance Quote Page — Custom Hooks and Context." You will move the form logic into a custom hook, share the saved-quotes list through context with `useReducer`, and remove prop drilling. The whole exercise takes about one hour.

In the lab you will:

- Open the `Labfiles/evergreen-quote-react-hooks` folder, run `npm install`, then `npm run dev`.
- Extract the form state and premium logic into a custom hook, `useQuoteEstimate`.
- Use the hook in `QuoteForm` so the component reads cleaner.
- Write the reducer and provider in `QuotesContext`, seeded with the sample quotes.
- Wrap App in the `QuotesProvider` and read the list from context in `RecentQuotes` (no props).
- Add a Save button that dispatches an add action and watch the new quote appear in a sibling.

The lab instructions (with "Why this matters" callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for students who want to go deeper beyond the core curriculum. They are outside the 4-hour weekly budget. Performance optimization and testing are covered next week in *React Beyond the Basics*, so focus this week on hooks, context, and reducers.

Topic	Material Description	Link / Reference
React	React Essential Training (Eve Porcello, LinkedIn Learning) — a complete course; the later chapters cover hooks, context, and project patterns in depth (Video Course).	linkedin.com/learning — React Essential Training
Reference	React.dev “Learn” — the official tutorial. The “Managing State” and “Escape Hatches” sections map directly to this week (Reference / Tutorial).	react.dev/learn
Reducers	Extracting State Logic into a Reducer — React.dev. A focused walkthrough of useReducer with a worked example (Tutorial).	react.dev/learn/extracting-state-logic-into-a-reducer
Hooks API	Built-in React Hooks — React.dev reference. The authoritative list and signatures for useState, useReducer, useContext, and more (Reference).	react.dev/reference/react/hooks
Testing	React Testing Library docs — the guiding principles and queries for testing components the way users interact with them (Reference).	testing-library.com/react